Analyzing and Visualizing Data Report

Energy Consumption Case Study

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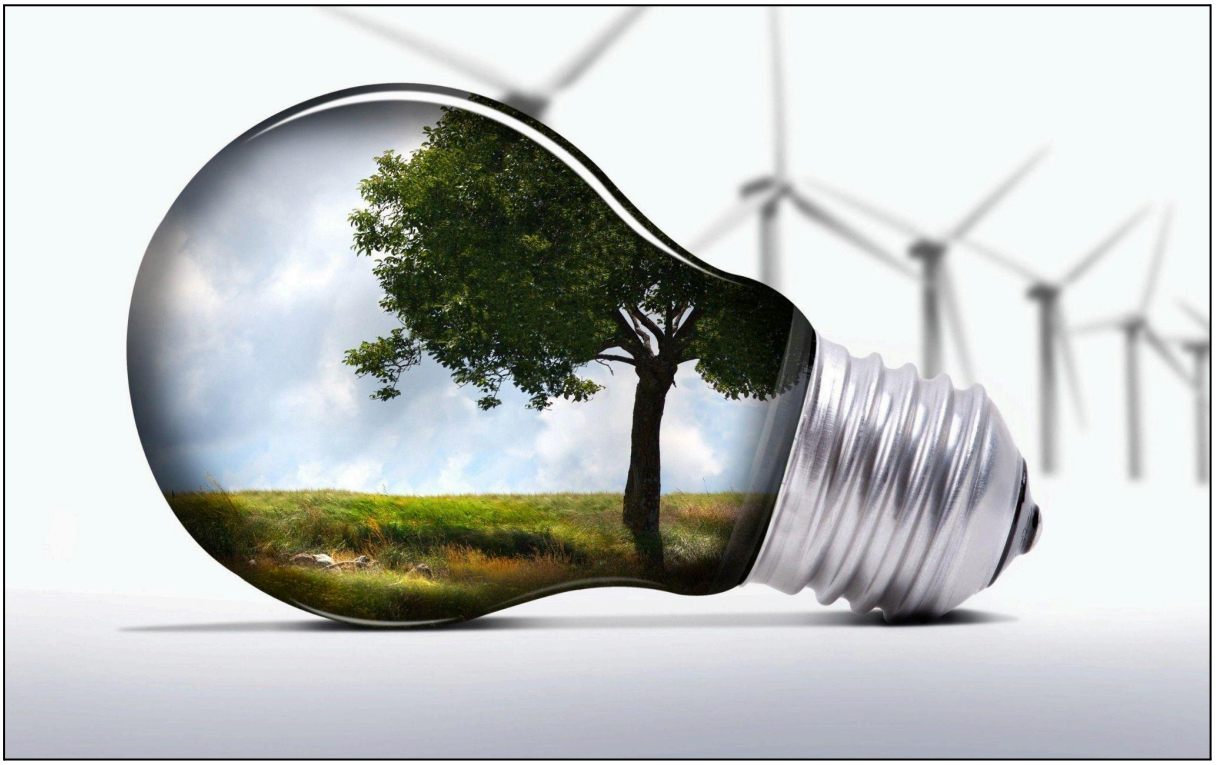

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CHAPTER 1 – BUSINESS / ORGANIZATION / PROBLEM SCENARIO

1.1. Presentation of the Organization and General Context

This Business Intelligence (BI) project focuses on the **detailed analysis of electricity consumption patterns in Portugal**, utilizing official data provided by the **E-REDES Open Data Portal**.

E-REDES, Distribuição de Eletricidade, S.A., is the **largest electricity distribution network operator in Portugal**, responsible for managing and operating approximately 230,000 km of distribution lines. As the concessionaire of the low, medium, and high voltage distribution network, E-REDES ensures the supply of electricity to more than 6 million customers, covering homes, businesses, and industries throughout the national territory.

E-REDES' main objectives focus on:

- Ensuring the quality and continuity of electricity distribution services
- Promoting the energy transition through the integration of renewable sources
- Optimizing the operational efficiency of the distribution network
- Supporting decarbonization and Portugal's sustainability goals

In the current context, Portugal intensified its efforts in the energy transition, with ambitious targets for 2030 that include a 55% reduction in greenhouse gas emissions (compared to 2005) and an increase in the share of renewable energy to 47% of final energy consumption. This framework emphasizes the importance of monitoring, analyzing, and predicting energy consumption patterns.

1.2. Informational Problem and Motivation for the BI Project

The management of modern electricity distribution networks has become increasingly complex due to the integration of decentralized generation and the growing component of renewable sources. These changes introduce new operational challenges, as consumption patterns have become more dynamic and are influenced by diverse economic, environmental, and social factors. At the same time, organizations face growing pressure to optimize

infrastructure investments, minimize losses, and meet ambitious sustainability targets that depend on a deeper understanding of **how energy is consumed across different regions and time periods**.

In this context, a central problem emerges: **E-REDES lacks integrated analytical tools that can provide a holistic view of network behavio**r. Existing data is fragmented and not easily transformed into insights that can reveal temporal trends, geographical disparities, or anomalies that may signal inefficiencies or risks.

The absence of such a unified analytical environment leads to **operational inefficiencies and reactive planning**. Resource allocation for maintenance and network expansion is often based on limited or delayed information, which impairs the identification of priority areas for intervention and the building of reliable medium- and long-term projections. This lack of predictive capacity reduces stakeholders' ability to anticipate emerging challenges, address local consumption surges, and design coherent strategies aligned with national and European energy transition goals.

To address these limitations, the BI project proposes the **development of an integrated analytical solution built on Microsoft Fabric, supported by semantic models and Power BI**. This platform will enable the organization of a robust **cloud-based data architecture that consolidates large volumes of energy consumption data in a secure and scalable environment**. Through interactive dashboards and multidimensional analysis tools, users will gain **clear and intuitive visibility over consumption patterns**, enabling them to **explore data across temporal and geographic dimensions**.

Beyond visualization, the solution aims to facilitate the identification of patterns, anomalies, and emerging trends that can **support proactive and data-driven decision-making**. The semantic modeling capabilities in Fabric will allow stakeholders to conduct ad-hoc analysis and answer specific business questions without requiring extensive technical knowledge. Furthermore, **by establishing a strong analytical foundation, the project prepares E-REDES and other actors for future predictive and prescriptive analyses**—such as **forecasting network demand or estimating the impact of policy interventions**. Overall, the project lays the groundwork for a more **informed, proactive, and sustainable energy distribution**.

The benefits of this initiative extend across the energy ecosystem. E-REDES and other key stakeholders can leverage the solution to enhance operational efficiency and **guide strategic infrastructure investments**. **Regulators and policymakers**, such as ERSE (Entidade Reguladora dos Serviços Energéticos), **gain access to more reliable data for market oversight and policy definition**. Municipalities obtain visibility into **local energy dynamics**, **supporting urban planning and sustainability initiatives**. Researchers and academic institutions can also use the structured database to **support studies in energy transition, consumption behavior, and network optimization**. All these parties will use this information to protect consumers and ensure a reliable, robust service.

1.3. Identification of Business Needs (BNs)

The following **business needs** have been identified to support effective energy consumption analysis and guide the development of the BI solution. They are **organized into**

major analytical categories, **each addressing a specific dimension of the project's informational requirements**. As the project evolves, these needs may be refined to incorporate new insights, datasets, or analytical priorities.

In addition, **anomaly detection remains a transversal requirement across all categories**, as identifying unusual or unexpected consumption behaviors is essential for diagnosing data quality issues, detecting operational anomalies, and uncovering opportunities for optimization or further investigation.

1.3.1. Time-Based Analysis

BN01 – Analysis of year-over-year consumption trends

Description: Evaluate how active energy consumption evolves, identifying **growth patterns**, declines, and **potential inflection points**.

- **Q:** What is the annual growth rate of energy consumption?
  DAX: *YoY Consumption*

- **Q:** Are there significant inflection points or changes in consumption trends across year partitions?
  DAX: *YoY Consumption by Month*, *YoY Consumption by Season*

BN02 – Identification of seasonal consumption patterns

Description: Detect **energy consumption seasonality** to anticipate demand variation and support resource planning.

- **Q:** Are there noticeable seasonal patterns in energy consumption?
  DAX: *Average Consumption Season*

- **Q:** How do seasonal patterns vary across regions?
  DAX: *Average Consumption Season by District/Municipality*

- **Q:** What is the difference between the highest and lowest consumption seasons (volatility)?
  DAX: *Seasonal Consumption Volatility*

BN03 – External factor influence on energy consumption

Description: Assess how **holidays** and **exceptional periods like the COVID-19 pandemic** influence energy consumption to improve forecasting and understand external consumption drivers.

- **Q:** Was energy consumption any different during COVID?
  DAX: *Average Consumption in COVID Period*

- **Q:** How do holidays impact consumption patterns?
  DAX: *Average Consumption by Holiday (e.g., July, August, December)*

1.3.2. Geographic Analysis

BN04 – Regional energy consumption variation across the country

Description: Evaluate how consumption varies **geographically characteristics** relate to consumption.

- **Q:** What are the total consumption values by District?
  DAX: *Total Consumption by District*

- **Q:** Which districts show the lowest/highest levels of consumption?
  DAX: *Minimum/Maximum Total or Average Consumption by District*

- **Q:** What is the proportion of energy consumption per Parish within a Municipality?
  DAX: *Average Consumption by Parish (within Municipality)*

- **Q:** Which regions show the greatest variability in consumption?
  DAX: *Consumption Amplitude by District / Consumption Volatility by District / Consumption Variation Coefficient by District*

- **Q:** Are there patterns of geographic concentration or decentralization?
  DAX: *Average Consumption by Municipality / District*

1.3.3. Voltage Level Analysis

BN05 – Relationship between voltage level and energy consumption

Description: Analyze how **consumption varies across voltage levels** (Low voltage versus Medium, High, and Very High voltage) to distinguish consumer types and identify operational implications.

- **Q:** How many parishes/municipalities use each voltage level?
  DAX: *Count of Parishes/Municipalities by Voltage Level*

- **Q:** How does the voltage level affect energy consumption?
  DAX: *Average Consumption by Voltage Level*

- **Q:** What is the total consumption across voltage levels?
  DAX: *Total Consumption by Voltage Level*

This chapter established the framework for the Business Intelligence project on energy consumption in Portugal, contextualizing E-REDES's role in the sector, identifying the central informational problem, and defining specific business needs organized into three analytical categories that will guide the development of the analytical solution.

These business needs directly address the exploratory data analysis questions outlined in the project scope, serving as the foundation for designing the data model, creating analytical

reports, and developing visualizations that transform raw data into actionable business intelligence.

CHAPTER 2 – DATA SOURCES

This chapter describes the data sources used in the project, the structure of the original dataset, and the ingestion and preparation steps implemented in Microsoft Fabric to support the analytical solution.

2.1. Datasets and Data Structure

The main dataset used in this project (**DS #5 – Energy Consumption**) is sourced from the E-REDES Open Data Portal, a platform that provides public information about electricity consumption in Portugal. This dataset contains monthly active energy consumption data (kWh) since November 2020 until July 2025, disaggregated by:

- Geographic dimension: District, Municipality, and Parish, with the respective codes;
- Infrastructure dimension: Voltage level (Low Voltage and Very High, High and Medium Voltage);
- Temporal dimension: Date, Year, and Month (along with Month int).

Some problems arose upon ingestion of the dataset to Microsoft Fabric. To ensure a proper import, special characters (spaces, underscores, and parentheses) were removed from the column names, as well as the comma present in the Voltage level column.

In addition to the main energy consumption dataset, the project incorporates a complementary geographic dataset (**base_geography.csv**) sourced from the SimpleMaps (World Cities Database) composed of:

- District and City names;
- Latitude and Longitude of each municipality.

Furthermore, an additional **Date Table** was ingested to facilitate temporal analysis and support the definition of hierarchical temporal structures (Year → Quarter → Month) that will be implemented later in the semantic model. The Date dimension ensures consistent time intelligence operations, enables accurate period comparisons, and standardizes all temporal relationships across the analytical solution. In addition, the Date Table was enriched with custom period indicators (COVID-19 period and holiday-related periods), enabling targeted analysis of consumption behavior during exceptional events and seasonal breaks, and supporting like-for-like comparisons with standard operating periods.

2.2. Workspace creation and Ingestion into Microsoft Fabric

All project data was organized within the "**AVD 2025 Group 07 Energy Consumption**" **workspace**, where all group members were added for collaborative development.

In this environment, the **Lakehouse "LH_SOURCES_ENERGY_CONSUMPTION"** was created and populated with the three datasets used in the project: the main energy consumption file (**Fact Consumption**), the geographic data (**Dim Geography**) obtained from SimpleMaps (World Cities Database), and the **Dim Date** table.

2.3. Transformations Applied (Dataflow Gen2)

The datasets were processed through **three Dataflow Gen2 pipelines** to ensure proper standardization and preparation before semantic modeling and reporting. Each dataflow was designed with a specific purpose, transforming raw source files into curated outputs that resulted in the final **dimension** and **fact** tables used in the Semantic Model.

Overall, the transformation process focused on ensuring:

- consistent column naming and formats;
- correct data types for analytical use;
- clean and reliable keys for relationships;
- creation of calculated/derived columns to support analysis and segmentation;
- removal of redundant or irrelevant fields.

All dataflows followed a common structure in Microsoft Fabric: Power Query connects to the project workspace, navigates to the Lakehouse where the raw tables are stored, applies the required transformations, and writes the cleaned and enriched outputs back to the Lakehouse for use in the Semantic Model.

DF #1 — Consumption Data Preparation (Fact table)

The purpose of this dataflow was to prepare the raw consumption dataset and generate the final **Fact Consumption** table used in the Semantic Model.

Main transformations applied:

- Column renaming and standardization (removal of special characters and harmonization of naming conventions);
- Data type validation and conversion (e.g., Date fields as date type, consumption values as numeric);
- Filtering of invalid records;
- Removal of redundant or non-analytical columns to reduce noise and improve model efficiency;
- Selection of relevant analytical fields required for time and geography analysis.

During the transformation of the consumption dataset, data quality issues were identified in the **Municipality Code** field, where some records contained invalid values stored as text instead of the expected numeric format. To ensure consistency and preserve the required structure of this key field, a custom transformation rule was implemented to automatically generate a **dummy numeric code** (using a predefined rule, such as appending "99" to the generated identifier) for

those invalid entries, following the same formating and filling rules applied to valid municipality codes. This step ensured clean and reliable keys for relationships in the semantic model.

Output: Fact Consumption

DF #2 — Geography Data Preparation (Dimension Table)

The purpose of this dataflow was to prepare and standardize the geographic dataset to generate the **Dim Geography** table used in the Semantic Model. This dimension supports territorial analysis and enables drill-down across geographic levels, as well as map visualizations through geographic coordinates of latitude and longitude.

This table contains the geographic information available in the original consumption dataset, including **districts and municipalities** and their respective **codes**. In addition, the dimension was enriched with **latitude and longitude at the municipality level**, enabling geographic analysis and supporting map-based visualizations in Power BI.

Main transformations applied:

- Column renaming and standardization (harmonization of naming conventions and removal of special characters);
- Removal of duplicate records to ensure a unique list of geographic entities;
- Selection of relevant attributes required for geographic analysis (e.g., District, Municipality, and geographic coordinates).

These transformations ensured that the geographic dimension provides clean and reliable keys for relationships with the fact table, supporting consistent filtering and correct aggregation behavior in Power BI.

Output: Dim Geography

DF #3 — Date Table Preparation (Dimension Table)

The purpose of this dataflow was to create and prepare the **Dim Date** table used in the Semantic Model. This dimension supports consistent temporal analysis and enables time intelligence calculations across the report (MoM, QoQ, and YoY comparisons), while also providing additional time-based segmentation for specific periods.

Main transformations applied:

- Generation of a complete calendar table covering the project's analysis period;
- Creation of standard calendar attributes required for reporting (Year, Quarter, Month, Month Name, and Month Number);
- Data type validation and formatting to ensure correct time-based filtering and relationships;
- Creation of two derived period indicators to support targeted analysis:
  - **COVID Period flag**, defined as the period between **Mar 2020 and Oct 2021**;
  - **Holiday Period flag**, defined as **July, August, and December**.

These additional columns enable the comparison of consumption behavior between regular periods and exceptional/seasonal periods, supporting more contextual and insightful analysis in Power BI.

Output: Dim Date

CHAPTER 3 – SEMANTIC MODEL

This chapter presents and describes the semantic model developed for the project, including the hierarchies, hidden columns, data types, formatting, summarization logic, and the calculated metrics implemented in Microsoft Fabric to support the analytical solution.

3.1. Dimensional Model Overview

The semantic model **SM_ENERGY_CONSUMPTION** was designed following a **Star Schema** approach to ensure analytical consistency, scalability, and performance. The model is centered on the **Fact Consumption** table, which contains monthly active energy consumption values (kWh) and the calculated measures used throughout the report.

The analysis is supported by two main dimensions:

- **Dim Date**, enabling consistent time-based analysis and time intelligence calculations (e.g., Month-over-Month, Quarter-over-Quarter, and Year-over-Year comparisons).
- **Dim Geography**, allowing consumption analysis at different territorial levels (District and Municipality), and supporting geographic visualizations through latitude and longitude attributes.

3.2. Table Relationships

The model relationships were implemented using a one-to-many structure, where each dimension filters the fact table:

- **Dim Date → Fact Consumption (one-to-many)**
  Ensures that all consumption measures can be analyzed by time, supporting both standard aggregation and time intelligence calculations.
- **Dim Geography → Fact Consumption (one-to-many)**
  Enables segmentation of consumption by district and municipality, supporting regional comparisons and spatial insights.

3.3. Hierarchies

Hierarchies were implemented to improve navigation and drill-down capabilities in the Power BI reports. The **time hierarchy** supports analysis from a yearly overview down to monthly detail, while the **geographic hierarchy** enables comparisons across different territorial levels (from District-level to Parish-level).

- **Time hierarchy:** Year → Quarter → Month
- **Geographic hierarchy (Summary):** District → Municipality
- **Geographic hierarchy (Detailed):** District → Municipality → Parish

3.4. Additional Label Tables (DAX Calculated Tables)

To improve report usability and readability, two additional label tables were created directly in the semantic model using DAX. These tables were introduced to provide descriptive category names for binary date flags (0/1), which are not user-friendly when displayed in slicers, legends, or axis categories.

Due to schema limitations when updating the Dim Date table through the Dataflow (new text columns were not consistently propagated to the existing destination table), the label logic was implemented at the semantic model level to ensure a stable and consistent structure for reporting.

The following calculated tables were created:

- **Dim_Covid_Label**: maps the `Is_Covid_Month` flag to meaningful labels (*COVID period* vs *Non-COVID period*).

- **Dim_Holiday_Label**: maps the `Is_Holiday_Month` flag to meaningful labels (e.g., *Holiday months* vs *Non-holiday months*).

Both tables were related to **Dim Date** through a one-to-many relationship using the corresponding binary flag columns. This approach preserves the original numeric fields for filtering and calculations, while enabling clearer category labels in visuals without altering the underlying fact/dimension schema.

3.5. DAX measures

The **Fact Consumption** table includes analytical Data Analysis Expression (DAX) measures to support the evaluation of temporal evolution and regional comparisons. Key metrics include average consumption, percentage variations (MoM, QoQ, and YoY), and variation/volatility/amplitude indicators, which are useful for identifying patterns and potential anomalies.

To improve model readability and ensure a more intuitive navigation experience in Power BI, the developed DAX measures were organized into **five functional categories**, according to their analytical purpose and contribution to the project's business questions. This approach simplifies metric selection during report development, reduces redundancy, and promotes clearer interpretation of the presented indicators.

01 - Main KPIs

- Total Consumption: *Total Consumption = SUM('Fact Consumption'[ActiveEnergy_kWh])*

- Average Consumption: *Average Consumption = AVERAGE('Fact Consumption'[ActiveEnergy_kWh])*

- Maximum Consumption: *Maximum Consumption = MAX('Fact Consumption'[ActiveEnergy_kWh])*

- Minimum Consumption: *Minimum Consumption = MIN('Fact Consumption'[ActiveEnergy_kWh])*

02 – Time Intelligence

- % MoM Avg Consumption: *DIVIDE([Average Consumption] - [LM Avg Energy Consumption], [LM Avg Energy Consumption], 0)*

- % QoQ Avg Consumption: *DIVIDE([Average Consumption] - [LQ Avg Energy Consumption], [LQ Avg Energy Consumption], 0)*

- % YoY Avg Consumption: *DIVIDE([Average Consumption] - [LY Avg Energy Consumption], [LY Avg Energy Consumption], 0)*

- % YoY Consumption: *DIVIDE([Total Consumption]-[LY Total Consumption],[LY Total Consumption])*

- LM Avg Energy Consumption: *CALCULATE([Average Consumption], DATEADD('Dim Date'[Date], -1, MONTH))*

- LQ Avg Energy Consumption: *CALCULATE([Average Consumption], DATEADD('Dim Date'[Date], -1, QUARTER))*

- LY Avg Energy Consumption: *CALCULATE([Average Consumption], DATEADD('Dim Date'[Date], -1, YEAR))*

- LY Total Consumption: *CALCULATE([Total Consumption],SAMEPERIODLASTYEAR('Dim Date'[Date]))*

03 – Share

- % of Total Consumption: *DIVIDE([Total Consumption],CALCULATE([Total Consumption], ALL('FACT Consumption])))*100*

- % Total Consumption District: *DIVIDE([Total Consumption],CALCULATE([Total Consumption], ALL('Dim Geography'[District])))*

04 – Geography

- Avg Consumption by District: *CALCULATE(AVERAGE('Fact Consumption'[ActiveEnergy_kWh]), ALLEXCEPT('Dim Geography','Dim Geography'[District]))*

- Total Consumption by District: *CALCULATE(SUM('Fact Consumption'[ActiveEnergy_kWh]), ALLEXCEPT('Dim Geography','Dim Geography'[District]))*

05 – Variability

- Consumption Amplitude: *[Maximum Consumption] - [Minimum Consumption]*

- Consumption Variation Coefficient: *DIVIDE(STDEV.P('Fact Consumption'[ActiveEnergy_kWh]),[Average Consumption]) * 100*

- Consumption Volatility: *STDEV.P('Fact Consumption'[ActiveEnergy_kWh])*

CHAPTER 4 – REPORT PRESENTATION AND DESCRIPTION

This chapter presents the analytical findings derived from the Power BI dashboards developed in Microsoft Fabric, addressing each Business Need identified in Chapter 1. The report is organized into sections corresponding to each business need, with detailed analysis of patterns, trends, and insights discovered through the data. Each page contains bookmarks to all the other sections (Fig. 1). Two custom tooltip graphs were added to show the individual municipalities within each district and the individual parishes within each municipality.

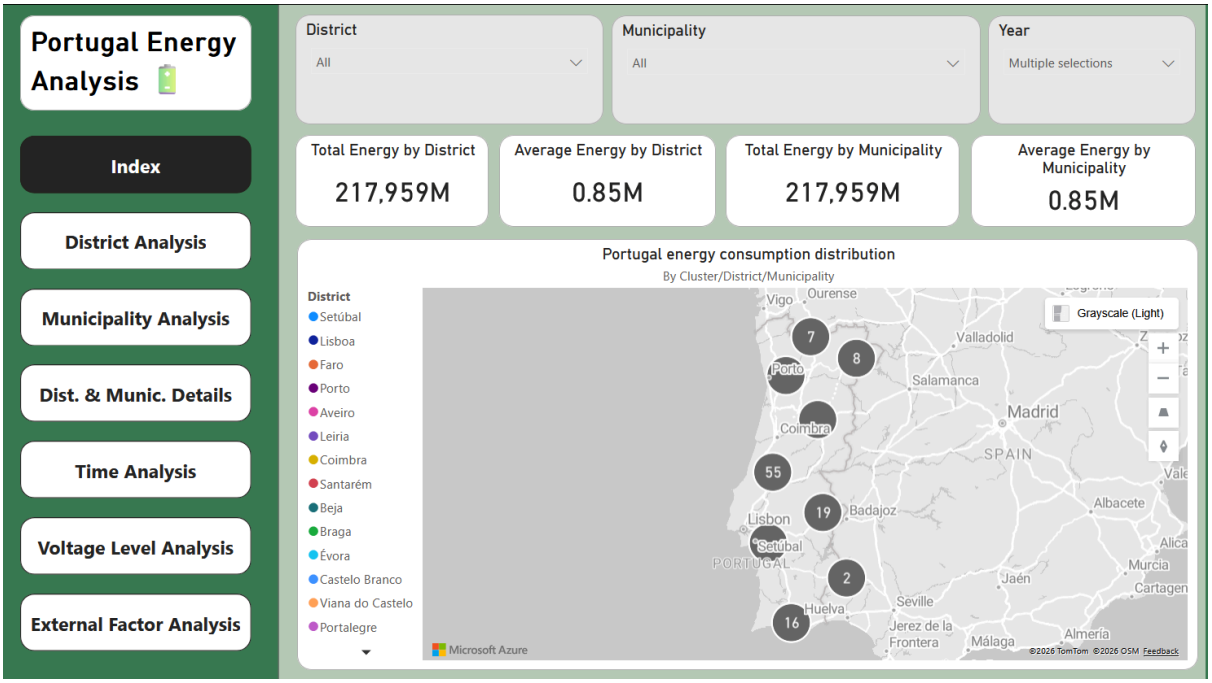


Figure 1. Index page of the PowerBI report, containing temporal and geographic slices and bookmarks to all other PowerBI report pages.

BN01 – Analysis of year-over-year consumption trends

The aim of this subsection was to address BN01 and evaluate how energy consumption evolves, identifying growth patterns, declines, and potential inflection points. To tackle this, we evaluated the annual growth rate of energy consumption and studied the changes in consumption trends across months and seasons (Fig 2).

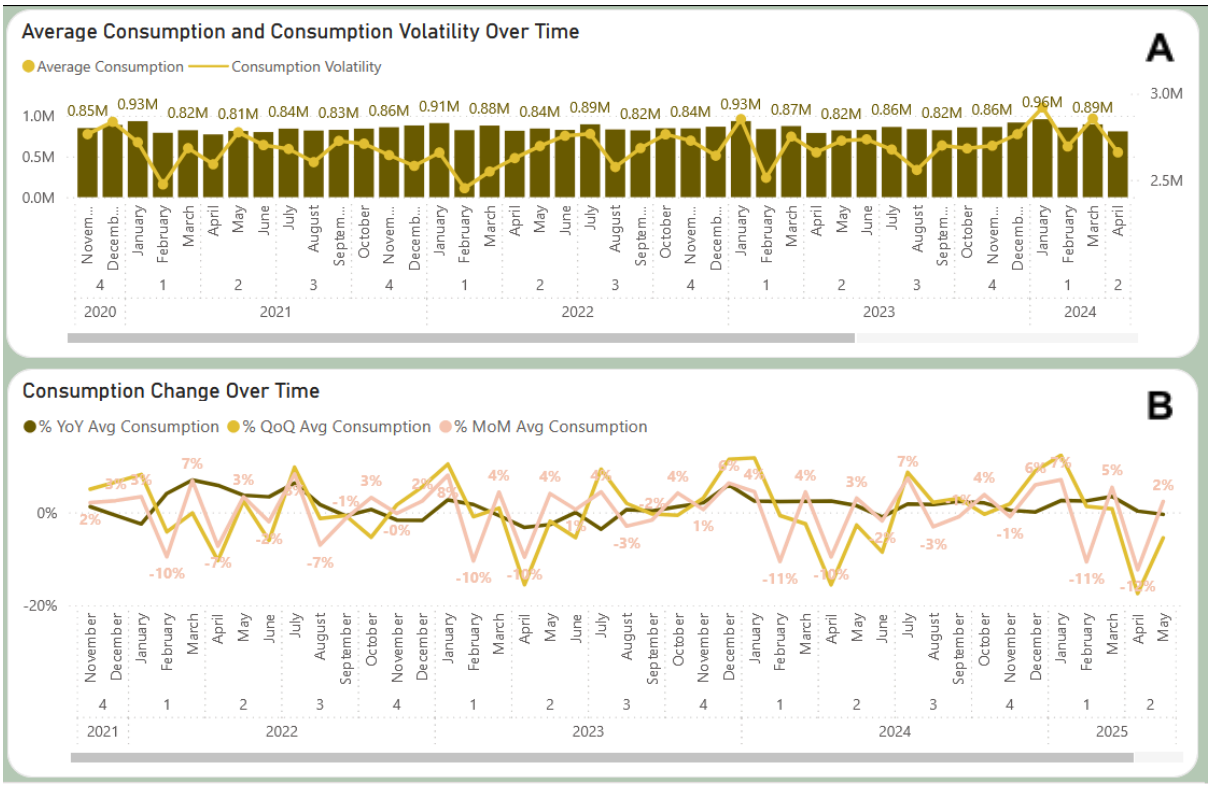


Figure 2. Analysis of nationwide (A) average consumption and consumption volatility over time, (B) year-over-year, quarter-over-quarter, and month-over-month consumption trends.

The year-over-year (YoY) consumption analysis (brown line in Fig. 2B) reveals stable growth rates oscillating around 0% from late 2021 to the first semester of 2025, indicating a mature electricity market where demand has plateaued. Fluctuations remain primarily within ±10%, with most periods comprising consumption between -5% and +5% compared to the previous year. This stability suggests economic expansion is offset by efficiency improvements, providing confidence for infrastructure planning based on historical consumption baselines rather than projecting continued growth.

The stabilization period from late 2021 through early 2023 shows all three evolution metrics (overall, monthly, seasonal) converging and maintaining relatively consistent patterns with limited volatility (Fig. 2). This convergence indicates a return to normal consumption dynamics where seasonal patterns and month-to-month variations follow predictable cycles rather than being disrupted by external shocks.

A major inflection point disrupts the otherwise stable trend, representing a period where consumption deviated from established patterns. Mid-2023 displays renewed volatility with sharp swings—positive peaks reaching +7% followed by significant drops to approximately -11% (Fig. 2A), likely reflecting combined effects of short-lived harsh weather anomalies, economic fluctuations, and behavioral shifts. During this inflection point, the three metrics (YoY, QoQ, MoM)

show divergent patterns: the monthly MoM exhibits the most extreme fluctuations, while the YoY and QoQ metrics display more moderate responses, highlighting how aggregation periods smooth short-term shocks (Fig. 2B).

Energy consumption per month is only comparable to the same period in other years, as variations along each year are very pronounced. In detail, there are energy consumption peaks during winter, and a dip in consumption during springtime.

The comparison between metrics reveals important dynamics: the monthly MoM (yellow-green line) shows more oscillation frequency with larger amplitude swings (Fig. 2B), reflecting sensitivity to short-term factors like weather variability and holiday patterns. The seasonal QoQ (yellow line) displays smoother but more sustained deviations, capturing mid-term trend changes (Fig. 2B). The YoY metric (brown line) provides the most stable baseline, filtering out seasonal effects (Fig. 2B). During normal periods, all metrics track closely around zero. During inflection points, however, they diverge substantially—the monthly metric shows more extreme peaks/troughs while the seasonal and yearly metrics display dampened but more sustained deviations. This divergence pattern signals exceptional circumstances requiring investigation.

BN02 – Identification of seasonal consumption patterns

The aim of this subsection was to address BN02 and detect energetic consumption seasonality to anticipate demand variation and support resource planning. To meet this need, we studied seasonal patterns over time, including the average consumption per season and its volatility (Fig. 3). Additionally, we analyzed geographic variations in consumption and year-over-year growth patterns across districts to understand where larger consumption variations are expected.

Clear seasonal patterns are visible in aggregate consumption data, with consistent annual cycles and, depending on the region, most districts display peaks during winter or summer months and troughs during spring, creating a somewhat predictable bimodal pattern (not shown). The consistency of these wave patterns year-over-year confirms that seasonal consumption is driven by heating/cooling demands tied to Portugal's Mediterranean climate.

Geographically (Fig. 3), consumption concentrates heavily in coastal urban districts. Lisbon leads with approximately 45bn kWh (~18% of total), followed by Porto (~36bn, ~15%), Setúbal (~24bn), and Aveiro (~20bn). These four districts collectively account for over half of national consumption. Interior districts (Guarda, Portalegre, Bragança) cluster at the bottom with significantly lower consumption levels.

Year-over-year growth patterns (2022-2024) reveal significant geographic heterogeneity in consumption dynamics:

<u>Growth leaders (2022):</u> Évora (+9%), Faro (+7%), and Castelo Branco (+6%) led growth, with Lisbon showing strong performance (+5%). Coimbra's decline (-3%) began in this year, foreshadowing 2023's sharper drop.

<u>Growth leaders (2023):</u> Beja (+6%), Faro (+4%), and Setúbal (+3%) experienced the strongest growth, with Faro's increase likely reflecting tourism sector expansion in the Algarve

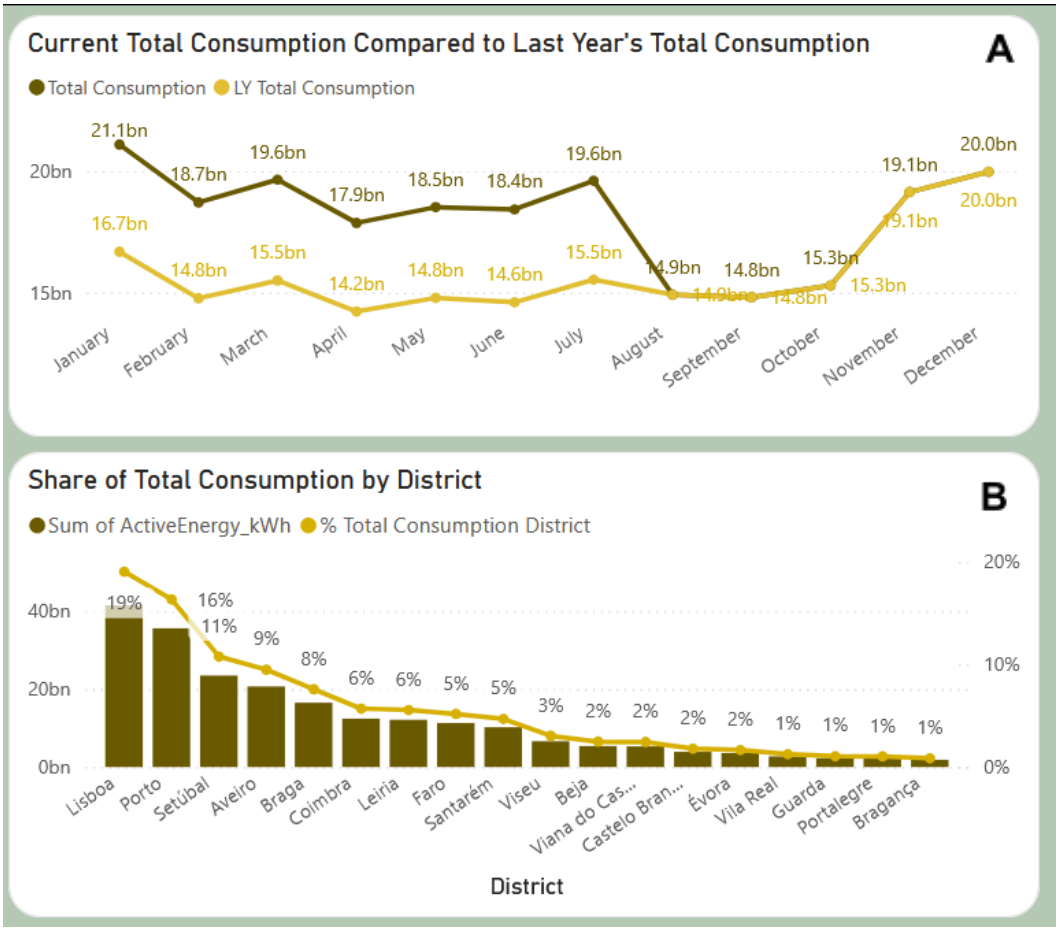
region. Lisbon and Porto, despite their large consumption bases, showed modest growth (+2% and +1% respectively).

Declining districts (2023): Coimbra experienced an accelerated decline (-9%, worsening from -3% in 2022), followed by Santarém (-2%) and several northern interior districts with -1% decreases (Leiria, Aveiro, Viseu).

Reversal patterns (2024): Notable shifts occurred in 2024, with Coimbra rebounding dramatically to +14% growth (the highest nationally, recovering from two consecutive years of decline). Conversely, previous growth leaders reversed: Beja from +7% to -4%, Évora from steady positive to -2%, and Faro's two-year growth streak (+7% in 2022, +4% in 2023) stalled at 0%, suggesting tourism-driven consumption may be stabilizing after post-pandemic expansion. Most districts converged toward modest positive growth (+2-3%), with broad-based increases across Viana do Castelo (+6%), Vila Real and Bragança (+3%), indicating a return to more uniform national patterns.

Geographic patterns: Coastal urban districts (Lisbon maintaining +2-5%, Porto +0-1%) show relatively stable moderate growth across all three years. Interior and southern districts display high volatility: Beja (+4% → +7% → -4%), Évora (+9% → +1% → -2%), Coimbra (-3% → -9% → +14%), and Portalegre (fluctuating around -1% to +3%). Tourism-dependent regions (Faro) demonstrate multi-year sensitivity to economic recovery cycles, with strong 2022-2023 growth followed by 2024 stabilization.

These three-year patterns reveal that high volatility districts require flexible capacity planning rather than linear projections, while stable coastal districts enable more predictable infrastructure investment strategies.



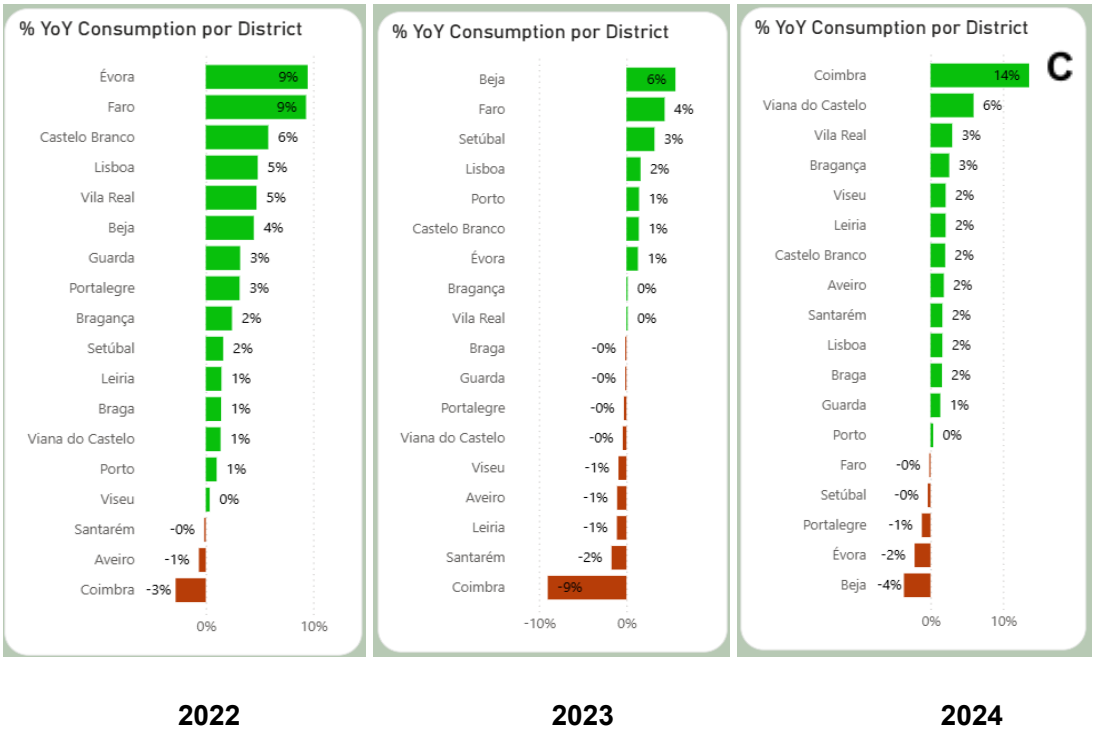


Figure 3. Evolution of consumption patterns over time and location. (A) Total consumption compared to last year's consumption. (B) Share of total consumption by district. (C) Year-over-year variation in 2022, 2023, and 2024 compared to the previous year.

BN03 – External factor influence on energy consumption

To answer BN03 and explore how external factors and exceptional periods can influence energy consumption, we chose two case studies: the COVID period and the holiday period. Energetic demand was slightly lower during COVID compared to the 'normal' period (Fig. 4A). Non-COVID months averaged ~0.86 M kWh while COVID months showed ~0.83 M kWh, representing approximately a 4% decline (Fig. 4A). Monthly (MoM) and seasonal (QoQ) metrics during the COVID period showed sharp negative deviations, reaching approximately -15 to -18% (not shown). This reflects the temporary cessation of commercial and industrial activities during lockdowns, reduced office occupancy, and changed mobility patterns. However, the modest aggregate decline masks sectoral variations—residential consumption actually increased as households adopted remote work, partially offsetting sharp declines in commercial sectors (retail, restaurants, offices). Industrial impacts varied by sector, with essential industries maintaining operations while discretionary manufacturing reduced output substantially.

Following the COVID-19 inflection point in early 2021, consumption trends display a recovery pattern with YoY metrics gradually returning to near-zero growth by late 2021 and into 2022. This recovery demonstrates the resilience of electricity demand and the temporary nature of pandemic-related consumption disruptions. By 2022, consumption patterns had largely normalized, with YoY growth rates stabilizing in the -5% to +5% range (Fig. 4A).

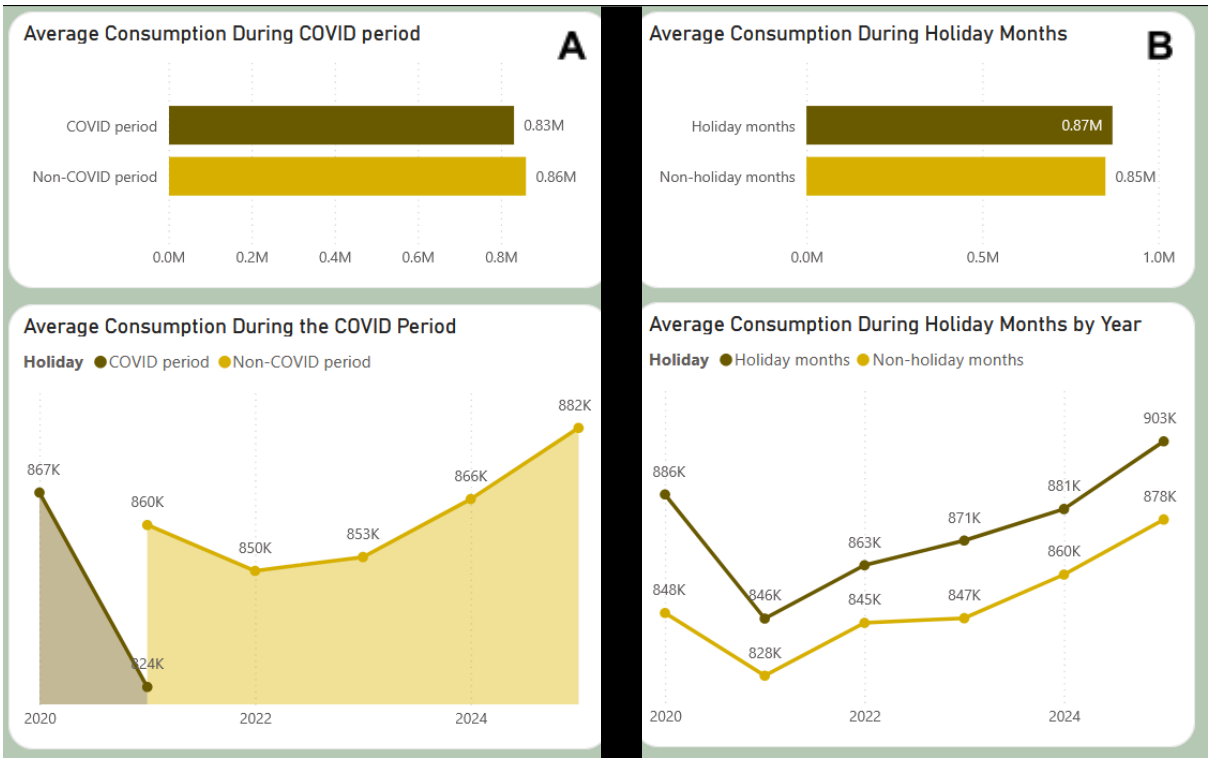


Figure 4. External factor influence on energy consumption patterns. Y-axis: Average consumption; X-axis: (A) During the COVID months (from March 2020 to September 2021) vs the Non-COVID period. (B) During holiday months July, August, and December, and outside holiday months.

During the holidays, the opposite was observed: there was actually an increased energetic consumption compared to the rest of the calendar year. The holiday months (July, August, December) averaged ~0.87 M kWh versus ~0.85 M for non-holiday months, representing approximately a 2.4% increase above baseline—contrary to expectations that reduced economic activity might decrease consumption (Fig. 4B). This counterintuitive finding reflects multiple reinforcing factors: these months coincide with peak heating (December) and cooling (July-August) demands driving residential consumption; tourism services intensify during summer in coastal regions (Algarve hotels, restaurants operating at maximum capacity); and extended household occupancy increases baseline residential usage (appliances, entertainment, climate control).

BN04 – Regional energy consumption variation across the country

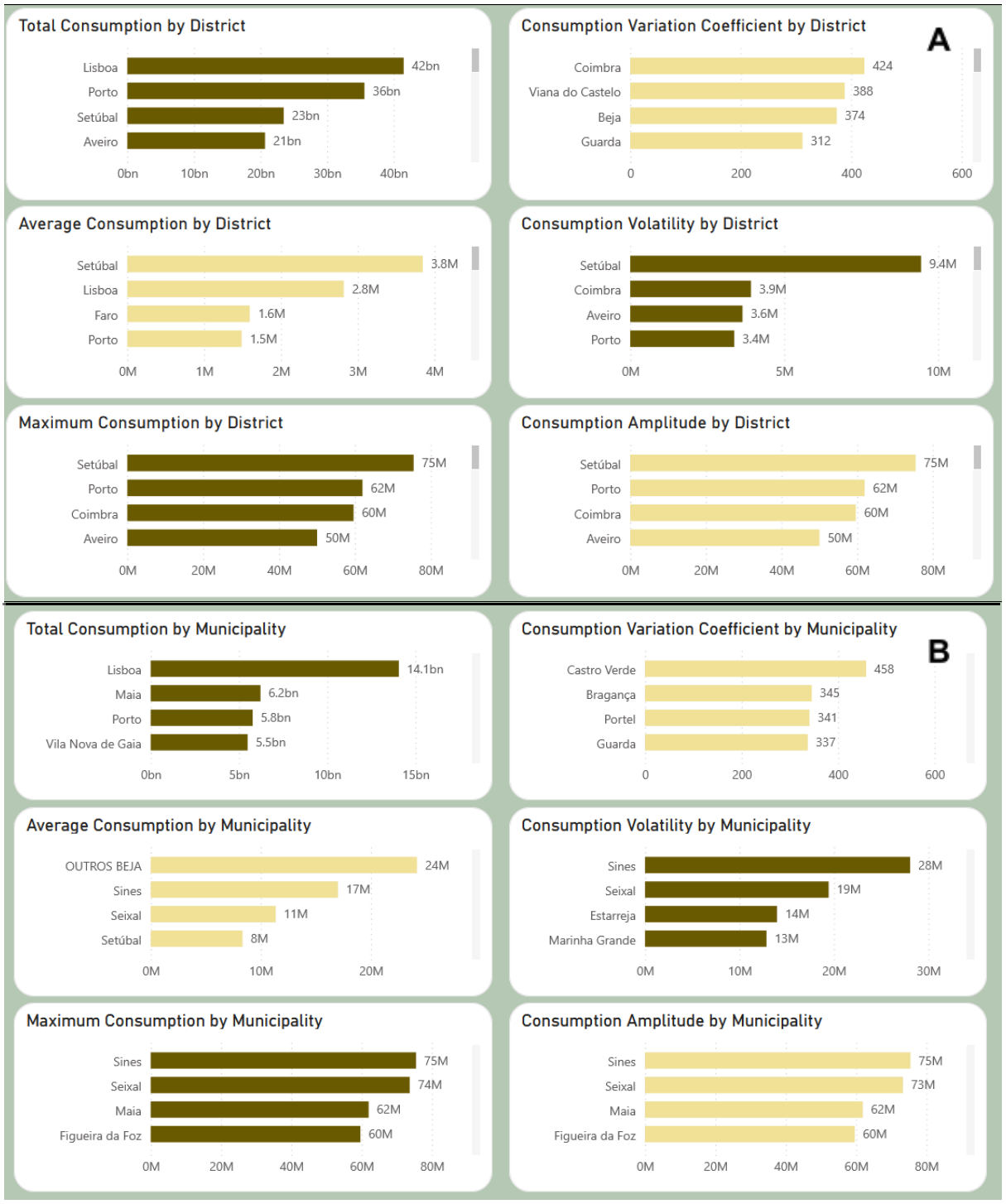


Figure 5. Regional energy consumption and variation metrics across the country's (A) Districts and (B) Municipalities.

To answer BN04 and evaluate how consumption varies geographically, we investigated consumption metrics by district and municipality (total, average, maximum, and minimum consumption) in the District and Municipality Analysis Dashboards (Fig. 5). Total energy consumption peaks in Lisbon (42 bn kWh), followed by Porto (36 bn kWh), Setúbal (23 bn kWh), and Aveiro (21 bn kWh) districts (Fig 5A). These four coastal/urban districts collectively account for ~56% of displayed consumption (122 bn out of ~220 bn kWh total). Mid-tier districts include

Braga (17 bn kWh), Coimbra (12 bn kWh), Leiria (12 bn kWh), Faro (11 bn kWh), and Santarém (10 bn kWh) (Fig. 5A). Interior districts (Viseu through Bragança) range from 2-7 bn kWh each (Fig. 5A), displaying a full order of magnitude less consumption than the most energetically costly districts.

When looking at maximum and minimum consumption (Fig. 5A), the district hierarchies changed order. Maximum consumption per district peaked in Setúbal (75 M kWh), followed by Porto (62 M), Coimbra (60 M), Aveiro (50 M), and Leiria (41 M). For average consumption, we observed Setúbal in the lead (3.8 M), followed by Lisbon (2.8 M), Faro (1.6 M), Porto (1.5 M), and Aveiro (1.4 M). Interior districts (Guarda, Bragança) show maximum values of 0.1-0.2 M kWh, again, an order of magnitude less than the districts that use more energy. Notably, Lisbon leads total but not maximum or average intensity, while Setúbal dominates both maximum and average metrics (Fig. 5A).

At the municipality level (Fig. 5B), the highest average consumption is concentrated in comparatively more industrialized municipalities. The aggregated category 'OUTROS BEJA' shows exceptionally high average consumption (24 M kWh), appearing as a clear outlier, followed by Sines (17 M), Seixal (11 M), and Setúbal (8 M). Notably, the Lisbon municipality, despite being the capital and having the highest total district consumption, does not appear among the top municipalities for average consumption (Fig. 5B), reflecting its diverse base with many consumption points but lower individual demand.

Coimbra, Viana do Castelo, and Beja had the highest variation coefficients (>350-400%) (Fig. 5A) . In terms of absolute volatility, Setúbal shows notably high standard deviation (9.4 M kWh), while Coimbra (3.9 M) and Aveiro (3.6 M) display more moderate volatility levels. Setúbal, Porto and Coimbra lead consumption amplitudes, with 60-80 M kWh swings between peaks and troughs (Fig. 5A). Notably, Faro displays a 40-50% seasonal amplitude, with major peaks during summertime.

Strong municipal-level geographic concentration patterns are visible in average consumption by Municipality and District. Industrial municipalities dominate: Sines, Seixal, Marinha Grande, Estarreja, and Matosinhos show the highest average consumption (Fig. 5B) due to energy-intensive facilities (refinery, steel, glass, chemicals, port operations). The 'OUTROS BEJA' aggregation exceeds 20 M kWh, appearing as an outlier (Fig. 5B). Lisbon, despite being the capital, appears relatively low on municipal ranking, reflecting its diverse base with more consumption points but lower per-point intensity. This demonstrates concentration in specific industrial municipalities rather than uniform distribution across urban areas.

BN05 – Relationship between voltage level and energy consumption

To answer BN05, energy consumption was analyzed by voltage level to assess spatial distribution, aggregate demand, and temporal variation (Fig. 6). Very High, High, and Medium Voltage supply reaches 294 municipalities and 1,603 parishes, while Low Voltage serves 280 municipalities and 2,699 parishes, with significant overlap as many areas contain both voltage infrastructures to serve different consumer types (Fig. 6C).

Very High, High, and Medium Voltage account for 113 bn kWh (52%) of total consumption, compared with 105 bn kWh (48%) for Low Voltage (Fig. 6A). Average consumption per entry is 1.24 M kWh for Very High, High, and Medium Voltage and 0.64 M kWh for Low Voltage, corresponding to an approximate 2:1 ratio (Fig. 6B). Despite this difference, total consumption remains relatively balanced across voltage levels.

Temporal analysis shows distinct seasonal patterns (Fig. 6D). Low Voltage consumption presents pronounced winter peaks (Fig. 6D), particularly in December and January, reaching approximately 2.5 bn kWh and exceeding higher voltage consumption during these periods (~2.0 bn kWh). In contrast, Very High, High, and Medium Voltage consumption remains comparatively stable throughout the year, with values ranging between 1.8 and 2.2 bn kWh and comparatively smaller peaks observed in July and October.

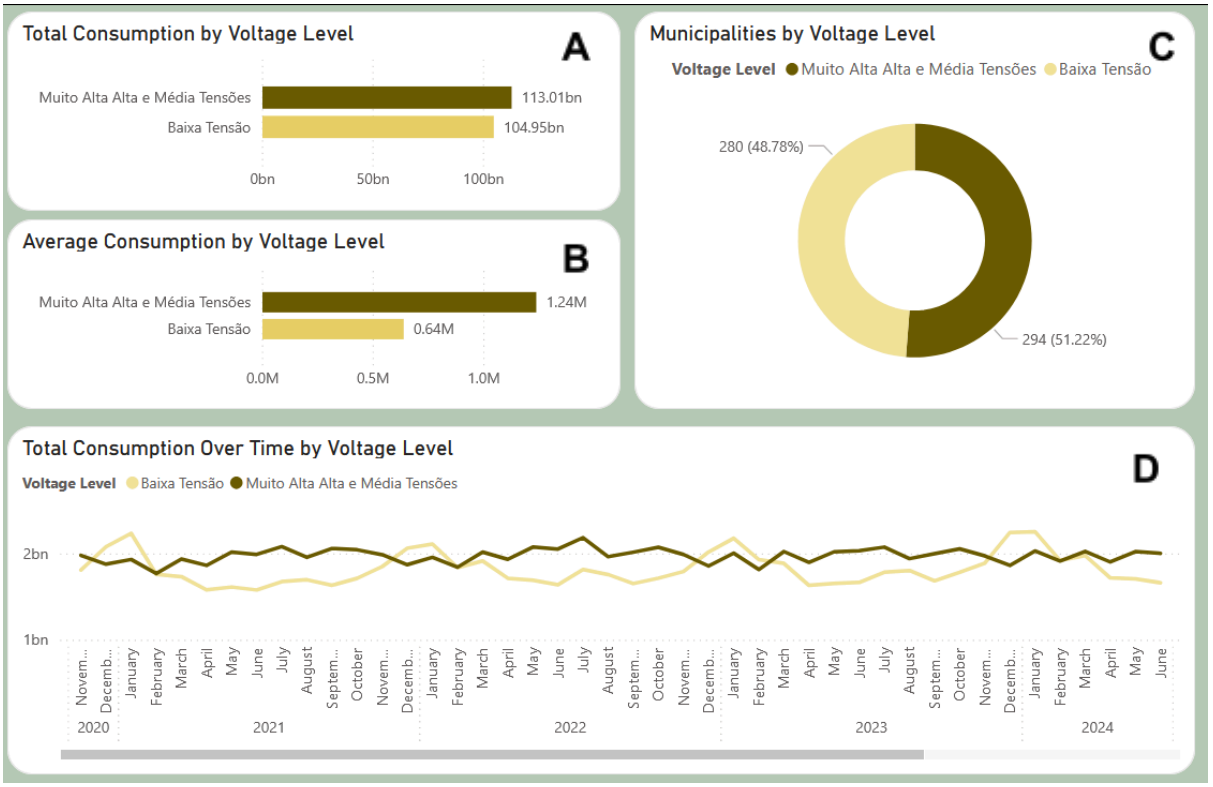


Figure 6. Relationship between voltage level and energy consumption. (A) Total Consumption by Voltage Level. (B) Average Consumption by Voltage Level. (C) Number of municipalities by Voltage Level. (D) Total Consumption Over Time by Voltage Level.

CHAPTER 5 – CONCLUSIONS / CRITICAL REVIEW

5.1. Key Findings

This analysis reveals a stable electricity consumption landscape in Portugal, clearly dependent on seasonal, geographic, and infrastructural patterns. At the national level, electricity demand displays a mature market profile, with year-over-year growth close to zero and consistent seasonal behavior, reflecting the balance between economic activity and efficiency improvements. The narrow band of variation (fluctuations primarily within ±10%) suggests that annual consumption changes are driven more by short-term factors (weather patterns, economic cycles, exceptional events) rather than sustained structural growth or decline trends. Consumption peaks occur in winter (December–January), driven primarily by residential heating demands, while spring months (April–May) present systematic troughs. Depending on the region, summer and winter periods both exhibit elevated demand due to increased cooling and heating needs, with holiday periods further amplifying approximately 2.4% consumption as residential occupancy rises without a full suspension of economic activity. The COVID-19 period resulted in a modest but measurable 3.5% consumption decline in demand, reflecting reduced industrial and service activity partially offset by increased residential use.

The quantitative similarity (both ~3% deviation) but opposite directionality (COVID decrease vs. holiday increase) demonstrates that external factors significantly influence consumption in ways standard time-series models might miss. Forecasting frameworks should incorporate scenario analysis for external shocks (pandemics, economic crises) and behavioral patterns (holidays, cultural events). The differential impact across consumer categories (residential vs. commercial/industrial) underscores the value of segmented forecasting approaches.

Geographically, electricity consumption is heavily concentrated in coastal and urban districts. Lisbon, Porto, Setúbal, and Aveiro together account for more than half of national consumption (approximately 56%), though different consumption metrics reveal distinct hierarchies. While Lisbon leads in cumulative consumption (approximately 45 bn kWh) due to population density, Setúbal consistently exhibits the highest maximum (75 M kWh) and average consumption intensity (3.8 M kWh), reflecting the presence of large industrial facilities such as the Sines refinery and Seixal steelworks.

Year-over-year growth patterns (2022-2024) reveal geographic heterogeneity and temporal volatility. Faro demonstrated strong tourism-linked growth in 2022 (+7%) and 2023 (+4%, surpassing Porto's +1%), though growth stabilized to 0% in 2024. Similarly, Beja showed high growth in 2023 (+7%) before reversing to -4% in 2024, while Évora declined from +9% (2022) to -2% (2024). Conversely, lower-demand interior districts such as Guarda, Bragança, and Portalegre display consumption levels an order of magnitude lower (2-7 bn kWh each) than Lisbon, reflecting both demographic and industrial composition effects. Notably, these districts show volatile year-over-year patterns, with Coimbra reversing from -3% (2022) and -9% (2023) to +14% (2024), demonstrating that consumption changes are not uniformly distributed. Coastal urban districts (Lisbon, Porto) maintain stable moderate growth (+2% and +0-1% respectively), while interior and southern districts display more volatile patterns. Industrial municipalities (Sines at 17 M, Seixal at 11 M average kWh) dominate per-facility consumption intensity, demonstrating that industrial composition influences demand more than population alone.

Infrastructure-level analysis further highlights structural differences in consumption behavior. Higher voltage consumers (Very High, High and Medium Voltage) exhibit double the average consumption levels of Low Voltage consumers (1.24 M vs 0.64 M kWh per connection point), reflecting continuous energy-demanding industrial operations. However, total consumption across voltage levels is nearly balanced (113 bn vs 105 bn kWh, or 52% vs 48%), as Low Voltage serves a far larger number of consumers nationwide (163.85 K vs 91.17 K connection points, representing 64.25% vs 35.75%).

Temporal dynamics differ significantly by voltage category: Low Voltage demand shows pronounced winter peaks (reaching approximately 2.3-2.5 bn kWh monthly) that surpass higher voltage consumption levels during cold months, whereas higher voltage demand remains relatively stable year-round (2.0-2.2 bn kWh) with minimal seasonal variation. This distinction implies that both voltage categories place comparable demands on generation and transmission capacity, while requiring fundamentally different distribution and operational strategies—Low Voltage requires capacity planning for sharp seasonal peaks driven by residential heating, while higher voltage infrastructure must support consistent high-intensity loads throughout the year.

5.2. Operational Recommendations for E-REDES

From an operational perspective, these findings support differentiated planning approaches for E-REDES. Capacity investments should prioritize high-demand districts while accounting for peak intensity rather than total consumption alone, particularly in industrial hubs such as Setúbal and seasonally volatile regions like Faro. Predictable spring demand troughs (April-May) provide optimal windows for maintenance activities, whereas winter and peak summer months require maximum network availability. Demand response strategies should be tailored by voltage level and regional profile, targeting residential heating Low Voltage loads in winter and tourism-driven cooling loads in summer, while industrial higher voltage programs should focus on continuous efficiency gains rather than seasonal peak management alone. E-REDES should also incorporate scenario analysis for external shocks (pandemics, economic crises) and behavioral patterns (holidays, tourism seasonality). The COVID-19 period demonstrated that aggregate stability masks significant sectoral variations, underscoring the value of segmented forecasting by consumer category (residential vs. commercial vs. industrial) rather than relying solely on aggregate models.

5.3. Limitations and Future Analysis

While the current analysis is constrained by the absence of weather variables, renewable generation data, and higher-resolution consumption records, it establishes a solid analytical foundation for understanding electricity demand patterns. Future work should prioritize increasing temporal granularity through daily or hourly data and incorporating key exogenous drivers such as weather conditions, resident population, and tourism intensity. These additions would enable a more nuanced interpretation of consumption dynamics, particularly during extreme weather periods, and allow demand to be more clearly linked to household and activity patterns.

Further analytical depth could be achieved by disaggregating consumption by voltage level, distinguishing low-voltage from high-voltage usage to better characterize residential versus industrial demand and improve regional comparability. With richer data, the framework could be expanded to include anomaly detection algorithms, real-time monitoring dashboards, and customer segmentation models. By defining expected consumption ranges across geographic levels and time periods, the system could automatically flag atypical behavior, supporting early identification of abnormal demand peaks, measurement errors, infrastructure issues, or shifts in local activity. Overall, this project delivers actionable insights into Portugal's electricity consumption and provides E-REDES with a scalable, extensible framework for enhanced planning, monitoring, and strategic decision-making, with significant potential for continued analytical development.

5.4. Methodological Remarks and Shortcomings

This project effectively supported our learning objectives and demonstrated the analytical potential of Microsoft Fabric and Power BI as integrated platforms for energy system analysis. Using these tools, we processed and visualized a large-scale temporal dataset of over 33 million records (2020–2025), enabling interactive exploration of electricity consumption patterns across time, geography, and infrastructure. The resulting dashboard and DAX measures support reproducible, multi-dimensional analysis and provide a solid basis for standardized, data-driven decision-making within E-REDES, while remaining extensible for future use.

At the same time, the project required overcoming several technical and methodological challenges. Data ingestion and preparation were affected by inconsistencies in column naming, categorical encoding, and regional Excel settings (different decimal separators, list delimiters, and date formats), all of which required systematic cleaning and harmonization to ensure reliable processing in Fabric. Data quality issues, such as aggregated municipality records lacking valid identifiers ("OUTROS"), necessitated the creation of surrogate codes to preserve analytical completeness without compromising the semantic model.

From a modeling perspective, differences in data granularity prevented the integration of population-based normalization metrics without introducing many-to-many relationships, unfortunately leading to the exclusion of demographic analyses from the final results, despite their relevance to the subject.

Report development also revealed limitations in Power BI visuals and aggregation behavior, requiring careful DAX adjustments to ensure correct results. In detail, certain card visual configurations did not return aggregated totals as expected, instead displaying the value of the first or last district in alphabetical order (e.g., Aveiro), rather than a true sum or average. Collaborative development was also constrained by Fabric's limited support for simultaneous editing, requiring careful coordination and sequential workflows among group members. Platform capacity constraints occasionally affected refresh times, causing temporary access issues, and occasional failures during dataflow execution and report updates.

The Microsoft Fabric version used during development did not include the standard “123” card visual commonly available in other Power BI versions, affecting the display of summary values and requiring the report to be restored to the default Power BI visual set to ensure correct and consistent aggregation behavior. The current Fabric version used in this project impacted development flexibility by not allowing SQL code execution directly within some transformation contexts, nor did it support the creation of new tables directly inside the semantic model in a fully stable way. As a workaround, certain classification and labeling logic (e.g., COVID and holiday labels) had to be implemented using DAX calculated tables instead of being incorporated upstream in the dataflows.

Despite these limitations, the project illustrates how robust and scalable BI solutions can be achieved through disciplined data preparation, thoughtful semantic modeling, and flexible use of Power BI and DAX. The challenges encountered strengthened the methodological rigor of the work and offered valuable insight into practical constraints typical of real-world, enterprise-scale BI environments.

REFERENCES

E-REDES – Distribuição de Eletricidade, “E-REDES Open Data Portal”. Accessed in “Data”. [Online] Available at https://e-redes.opendatasoft.com/pages/homepage/